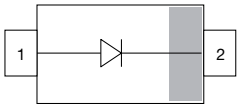
Small Signal Fast Switching Diode





FEATURES

- Silicon epitaxial planar diode
- Fast switching diode
- AEC-Q101 qualified available
  (part number on request)
- Molding compound meets UL 94 V-0
  flammability rating
- Moisture sensitivity level (MSL) 1
- Base P/N-G3-green, commercial grade
- Material categorization:
  for definitions of compliance please see
  www.vishay.com/doc?99912

Pb-free

e3

RoHS COMPLIANT

HALOGEN FREE

GREEN (5-2008)

LINKS TO ADDITIONAL RESOURCES

3D Models | Models | Marking | Parametric Search | Order Samples

MECHANICAL DATA

Case: SOD-123
Weight: approx. 10.6 mg
Packaging codes / options:
18/10K per 13" reel (8 mm tape), 10K/box
08/3K per 7" reel (8 mm tape), 15K/box

PARTS TABLE

PART	ORDERING CODE	AEC-Q101 QUALIFIED	TYPE MARKING	CIRCUIT CONFIGURATION	TAPED UNITS PER REEL	MINIMUM ORDER QUANTITY
1N4448W-G	1N4448W-G3-08	no	AJ	Single	3000 (8 mm tape on 7" reel)	15 000
	1N4448W-G3-18	no			10 000 (8 mm tape on 13" reel)	10 000

PACKAGE

PACKAGE NAME	WEIGHT	MOLDING COMPOUND FLAMMABILITY RATING	MOISTURE SENSITIVITY LEVEL	SOLDERING CONDITIONS
SOD -123	10.6 mg	UL 94 V-0	MSL 1 (according J-STD-020)	Peak temperature max. 260°C

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Reverse voltage		V_R	75	V
Repetitive peak reverse voltage		V_{RRM}	100	V
Continuous froward current (1)		I_F	300	mA
Average rectified current half wave rectification with resistive load (1)	f ≥ 50 Hz	$I_{F(AV)}$	250	mA
Surge current (1)	t < 1 s and T_j = 25 °C	I_{FSM}	500	mA
Power dissipation (1)	On FR-4 board with recommended soldering footprint	P_{tot}	280	mW
	Infinite heatsink		380	mW

Note
(1) Infinite heatsink

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THERMAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	**TEST CONDITION**	**SYMBOL**	**VALUE**	**UNIT**
Thermal resistance junction to ambient air	According to JEDEC® 51-3 on FR-4 board with recommended soldering footprint	R_{thJA}	440	K/W
Thermal resistance junction to lead	Infinite heatsink	R_{thJL}	330	K/W
Junction temperature		T_j	150	°C
Storage temperature		T_{stg}	-65 to +150	°C
Operating temperature		T_{op}	-55 to +150	°C

ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)						
PARAMETER	**TEST CONDITION**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Forward voltage	I_F = 100 mA	V_F			1	V
	I_F = 5 mA	V_F	0.62		0.72	V
Leakage current	V_R = 20 V	I_R			25	nA
	V_R = 75 V	I_R			2	µA
	V_R = 20 V, T_J = 150 °C	I_R			50	µA
Capacitance	V_F = V_R = 0 V				1.5	pF
Reverse recovery time	I_F = 10 mA, i_R = 1 mA, V_R = 6 V, R_L = 100 Ω	t_{rr}			4	ns

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TYPICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

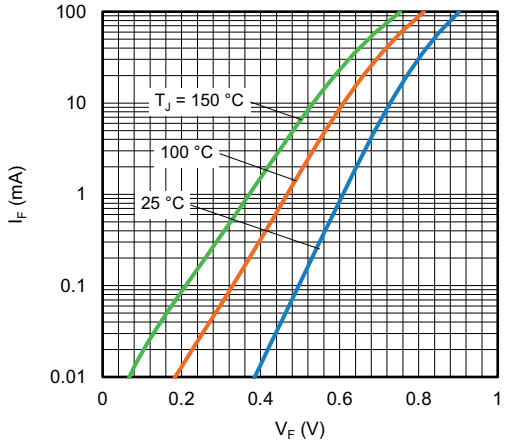


Fig. 1 - Typical Forward Current vs. Forward Voltage

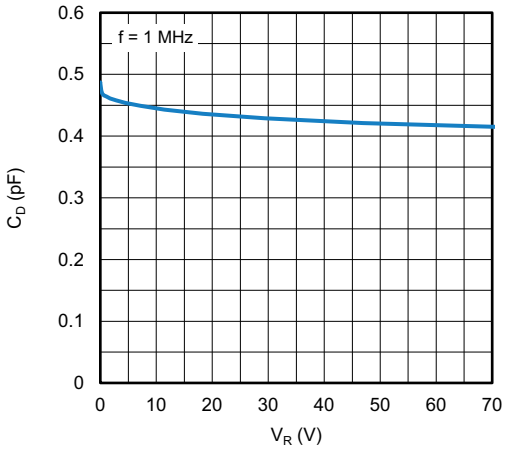


Fig. 3 - Typical Capacitance vs. Reverse Voltage

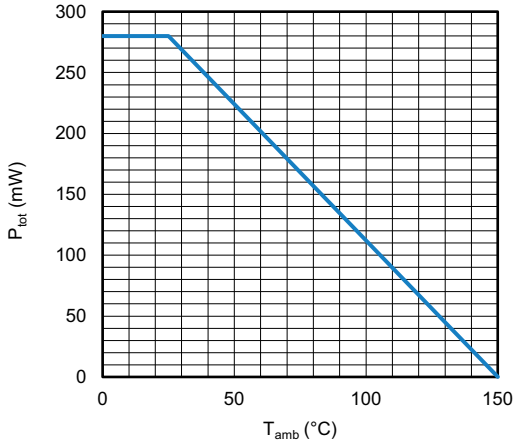


Fig. 2 - Admissible Power Dissipation vs. Ambient Temperature

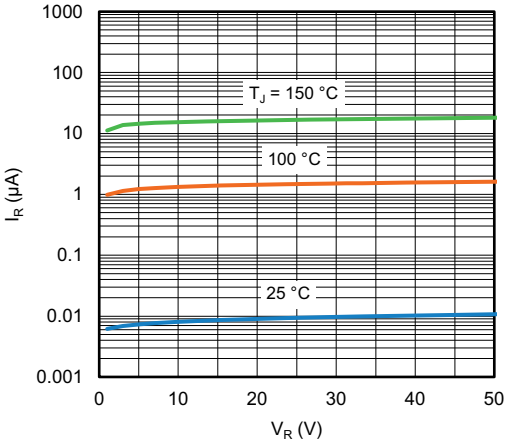
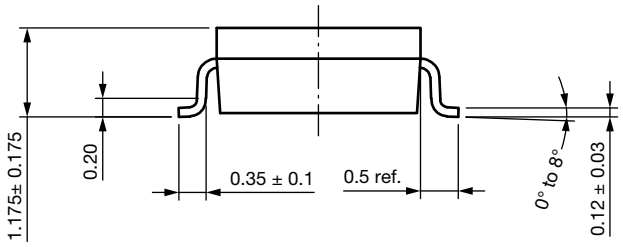
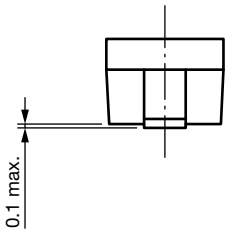


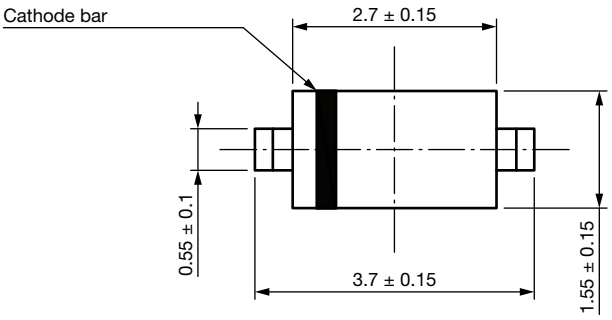
Fig. 4 - Typical Capacitance vs. Reverse Voltage

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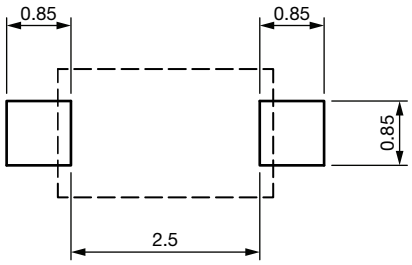
PACKAGE DIMENSIONS in millimeters (inches): **SOD-123**







Foot print recommendation

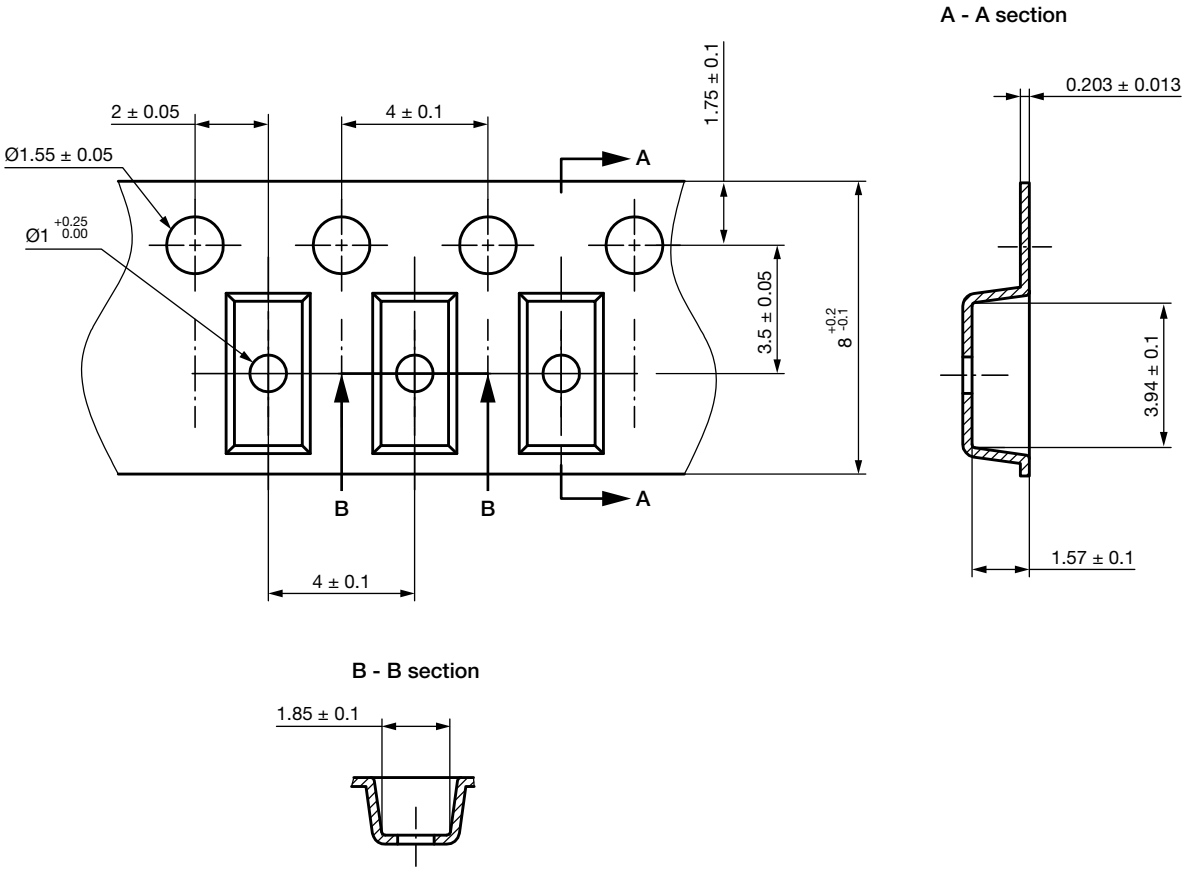


Rev. 01 - Date: 18. Jan. 2022
Document no.: S8-V-3910.01-003 (4)

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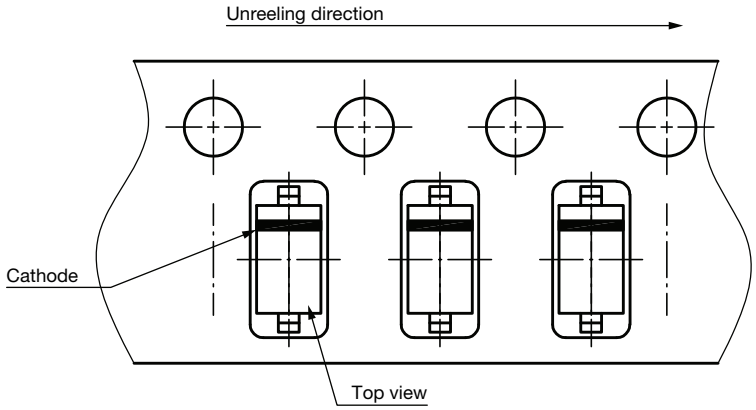
CARRIER TAPE SOD-123



Rev. 02 - Date: 21. Jan. 2014
Document no.: S8-V-3717.10-002 (4)

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ORIENTATION IN CARRIER TAPE SOD-123



Rev. 02 - Date: 07. Nov. 2022
Document no.: S8-V-3717.10-003 (4)

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